

Answer Key & Teacher Notes

Session 1 – Rational Numbers on a Number Line • Session 2 – Opposites of Rational Numbers

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Rational Numbers on a Number Line

TALK PROMPTS

- How do you decide how many equal parts to mark between two integers?
- Why is -3.5 to the left of -3 , even though 3.5 is greater than 3 ?

PARTNER TASK

Lab Thermometer. One partner plots -2.5 and 2.5 on a -4 to 4 number line marked in halves; the other proves they are opposites using distance from 0. Switch roles with -1.25 and 1.25 .

WATCH FOR

- Counting tenths to the right of -3 to plot -3.4 (the point belongs between -4 and -3).
- Splitting the gap by the numerator ($7/4$: split into 4 parts, not 7).
- Thinking a whole number such as 7 is not rational.

SESSION 2 • TEACHER NOTES

Opposites of Rational Numbers

TALK PROMPTS

- Why does taking the opposite twice bring a number back to where it started?
- What does the sign in $-(-4.5)$ mean in words?

PARTNER TASK

Card Flips. One partner writes a fraction or decimal on a card; the other names its opposite and its distance from 0. Flip again and predict the result before checking on a number line.

WATCH FOR

- Believing the opposite of a number is always negative.
- Reading $-(-a)$ as “more negative” instead of “the opposite of $-a$.”
- Thinking 0 has no opposite (it is its own opposite).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.C.6c	1	2	1	1	2	2	2	2	3	1	1
S1 • Form B	6.NS.C.6c	1	2	1	1	2	2	2	2	3	1	1
S2 • Form A	6.NS.C.6a	1	2	2	1	2	2	1	2	3	1	1
S2 • Form B	6.NS.C.6a	1	2	2	1	2	2	1	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

rational number: A number that can be written as a **fraction** of two integers.

integer: A whole number or its **opposite**.

number line: A line where numbers **increase** from left to right.

- To plot $7/4$, split each whole into **4** equal parts and count **7** parts from 0.
- Negative numbers sit to the **left** of 0 on a horizontal line and **below** 0 on a vertical line.

Example: -2.75 is between the integers **-3** and **-2**. Split the gap between -3 and -2 into **4** equal parts (fourths). Start at -2 and count **3** fourths to the left. Answer: -2.75 is one fourth to the **right** of -3 .

YOU TRY

1. -1.5 halfway between -2 and -1 ; 2.25 one fourth to the right of 2
2. a. -4 and -3 b. 2 and 3

EXIT TICKET

1. a. -5 and -4 b. 2 and 3
2. -0.75 three fourths left of 0; 1.5 halfway between 1 and 2
3. **B** (-2.25)

PRACTICE FORM A

1. a. -5 and -4 b. 0 and 1 c. -3 and -2 d. -3 and -2
2. a. 3.5 , -0.75 , 0.5 , -0.2 b. 3.5 halfway between 3 and 4; -0.75 three fourths left of 0; 0.5 halfway between 0 and 1; -0.2 two tenths left of 0
3. 2.5 degrees (both readings are 2.5 from 0)
4. -1.25
5. $7/1$; Lena is not right, so 7 is rational
6. -0.45 ($-11/4 = -2.75$ is between -3 and -2 ; 1.3 is between 1 and 2)
7. $1.6 - 0.4 = 1.2$ m
8. a. -12.5 m b. -13 and -12
9. Counting right of -3 moves toward 0, which gives -2.6 . -3.4 is 0.4 to the LEFT of -3 , so it is between -4 and -3 .
10. **B** (-3 and -2)
11. **A** (-4.5)

Self-check word NUMBER

PRACTICE FORM B

1. a. -4 and -3 b. 0 and 1 c. -3 and -2 d. -4 and -3
2. a. 2.5 , -0.5 , 1.25 , -0.3 b. 2.5 halfway between 2 and 3; -0.5 halfway between -1 and 0; 1.25 one fourth past 1; -0.3 three tenths left of 0
3. 3.5 degrees (both readings are 3.5 from 0)
4. -1.75
5. $-4/1$; Jalen is not right, so -4 is rational
6. -0.35 ($-9/4 = -2.25$ is between -3 and -2 ; 1.6 is between 1 and 2)
7. $2.1 - 0.6 = 1.5$ m
8. a. -8.5 m b. -9 and -8
9. Counting right of -3 moves toward 0, which gives -2.4 . -3.6 is 0.6 to the LEFT of -3 , so it is between -4 and -3 .
10. **C** (-2 and -1)
11. **C** (0.25)

Self-check word POINTS

GUIDED NOTES

opposites: Two numbers the same **distance** from 0, on different sides of 0.

opposite of the opposite: Taking the opposite **twice** gives back the number you started with.

zero: The only number that is its **own** opposite.

- The symbol $-(4.5)$ means the **opposite** of 4.5, which is -4.5 .
- The opposite of a negative number is **positive**.

Example: -5.3 is **5.3** units to the left of 0. Its opposite is the same distance on the **right** side of 0. So $-(-5.3) = 5.3$. Answer: $-(-5.3) = 5.3$

YOU TRY

- a. $-2/9$ b. 0.6 c. 0
- B = 1.25

EXIT TICKET

- a. 3.2 b. $7/9$ c. -1.4
- -6.5
- No. The opposite of -4 is 4, which is positive.

PRACTICE FORM A

- a. 2.5 b. $-7/10$ c. $1/4$ d. 0 e. -4.5 f. 4.5
- a. 4.5, 7.25, 11.5, 0.8 b. 7.25 m
- $-7/8$ (first flip $7/8$, second flip $-7/8$)
- 18.6 dollars
- 4.8
- a. -6.75 ft b. 6.75 ft each
- -6.35
- a. 4.5, -4.5 , 0.5 b. -0.5
- The first flip gives -8.6 ; the second flip goes back across 0 to 8.6. Two opposites return to the start, so the answer is 8.6.
- B** (3.8)
- C** ($5/6$)

Self-check word **REFLECT**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • RATIONAL NUMBERS ON A NUMBER LINE

- A = -1.75 ($-1\frac{3}{4}$) B = -0.5 ($-1/2$) C = 1.25 ($1\frac{1}{4}$)
- -2.5 halfway between -3 and -2 ; $1/2$ halfway between 0 and 1; -0.75 three fourths left of 0; $1\frac{1}{4}$ one fourth right of 1.
- a. -5 and -4 b. 3 and 4 c. -1 and 0 d. -3 and -2
- a. -3.50 b. $2\frac{1}{2}$ c. -4.5
- a. frog 1.5, fish -2.25 b. the frog (1.5 units from 0 is less than 2.25 units from 0)
- -1.5 is 1.5 units left of 0, so it is past -1 . It belongs halfway between -2 and -1 .

PRACTICE FORM B

- a. 3.5 b. $-3/10$ c. $3/4$ d. 0 e. -6.2 f. 6.2
- a. 3.5, 6.25, 10.5, 1.2 b. 6.25 m
- $-5/8$ (first flip $5/8$, second flip $-5/8$)
- 24.3 dollars
- 3.6
- a. -5.25 ft b. 5.25 ft each
- -8.15
- a. 3.5, -3.5 , 0.25 b. -0.25
- $-(-7.4)$ means the opposite of -7.4 . The opposite of a negative number is positive, so $-(-7.4) = 7.4$.
- B** (2.6)
- C** ($3/7$)

Self-check word **FLIPPED**

SHEET 2 • OPPOSITES OF RATIONAL NUMBERS

- a. -4.8 b. $2/3$ c. 0.05 d. 0 e. 1.6 f. $5/8$
- Q at 2.25 ($2\frac{1}{4}$), one fourth to the right of 2.
- 12.5 points; it is 12.5 points from 0 (same distance as -12.5)
- a. M = -3.75 ($-3\frac{3}{4}$) b. N = 3.75 ($3\frac{3}{4}$)
- $-(-2/5)$ means the opposite of $-2/5$. The opposite of a negative number is positive, so $-(-2/5) = 2/5$.
- B** ($7/8$)
- C** (3.4)